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Mitigating Hallucinations in Multi-modal Large Language Models via Image Token Attention-Guided Decoding

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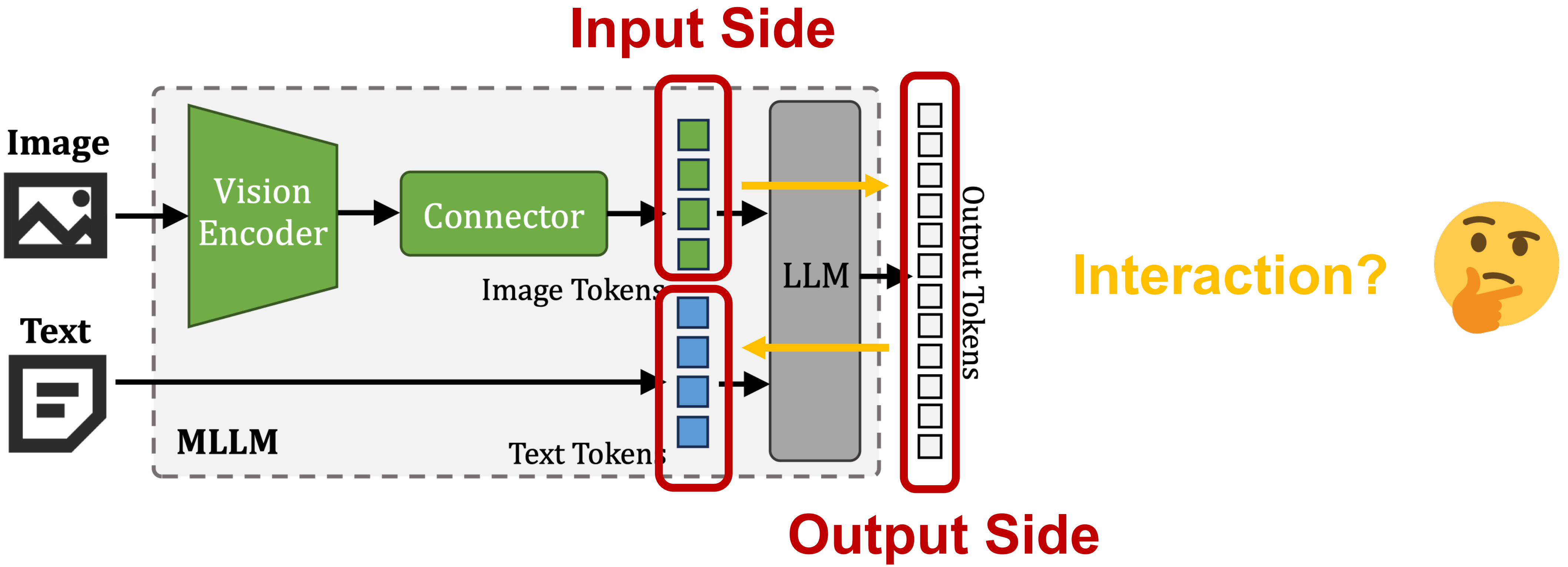
Mitigating Hallucinations in MLLMs

A New Trend:

Training-free methods that leverage the inherent characteristics of MLLMs



Interaction-Based Hallucination Patterns

An Empirical Result on Output Token's Attention Weights:

The hallucinatory segments in MLLMs are correlated with the reduction of attention weights to image tokens.

Segments	LLaVA-1.5	InstructBLIP	MiniGPT-4	mPLUG-Owl
w/o *halluc.*	12.0	79.2	40.2	59.8
w/ *halluc.*	10.9	74.8	37.7	56.0

The Average Attention Weights to Image Tokens

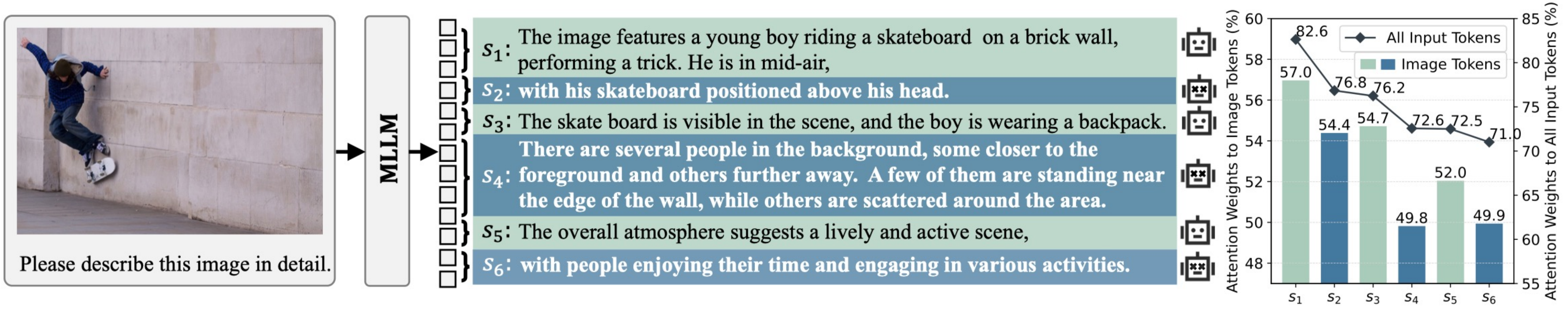
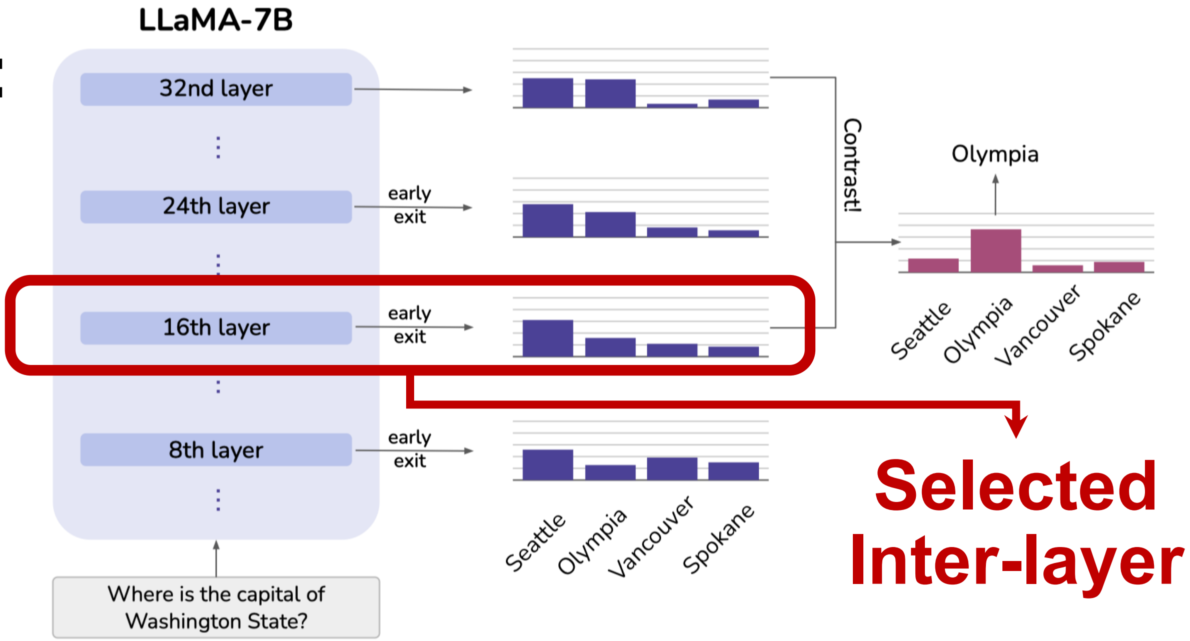


Image Token Attention-Guided Decoding

Inter-layer Contrastive Decoding:

Amplifying factual knowledge in LMs by exploiting modular encoding through contrastive decoding.



Intuitive Ideas:

- **Quantify** the difference in output tokens' attention to image tokens across different decoding layers.
- To best highlight the progression of image understanding, the selected intermediate layer should have **the most distinct** attention to image tokens compared to the last layer.

Chuang Y S, Xie Y, et al. Dola: Decoding by contrasting layers improves factuality in large language models, 2024

Image Token Attention-Guided Decoding

Image Token Attention-Guided Decoding (iTaD)

- Proposing an **image token attention vector** to measure the inter-layer difference in image understanding.
- Introducing a novel intermediate layer selection strategy for inter-layer contrastive decoding to mitigate hallucinations in MLLMs.

Features

- Leveraging MLLMs' internal representations and exploiting the attention to image tokens to mitigate their hallucinations.
- A training-free and plug and play method with minimal additional inference cost.

iTaD: An Implementation

Inter-layer Contrastive Decoding Process

The last layer

$$p_N(x_{t+1}|x_{<t+1}) = \texttt{softmax}(\texttt{Proj}(h_t^N))$$

Intermediate layer

$$p_M(x_{t+1}|x_{<t+1}) = \texttt{softmax}(\texttt{Proj}(h_t^M))$$

Hidden state for n^{th} layer:

$$h_t^n = \mathrm{MLP}(\mathrm{MHA}(h_t^{n-1})), \quad n = 1, 2, \ldots, N$$

$$\hat{p}(x_{t+1}|x_{<t+1}) = \texttt{softmax}(\mathbb{I}(x_{t+1}) \cdot \log \frac{p_N(x_{t+1}|x_{<t+1})}{p_M(x_{t+1}|x_{<t+1})})$$

Contrastive decoding

Constraint Function to Avoid False Positive Cases

$$\mathbb{I}(x_{t+1}) = \begin{cases} 1 & \text{if } x_{t+1} \in \mathcal{C}_{t+1} \\ -\infty & \text{otherwise,} \end{cases} \qquad \mathcal{C}_{t+1} = \{x_{t+1} \in \mathcal{V} : p_N(x_{t+1}) \geq \alpha \max_{x'_{t+1} \in \mathcal{V}} p_N(x'_{t+1})\}$$

iTaD: An Implementation

Constructing Image Token Attention Vector

Selecting the maximum weight in multi-head attention, which usually indicates the strong confidence of models.

Attention weights of MHA:

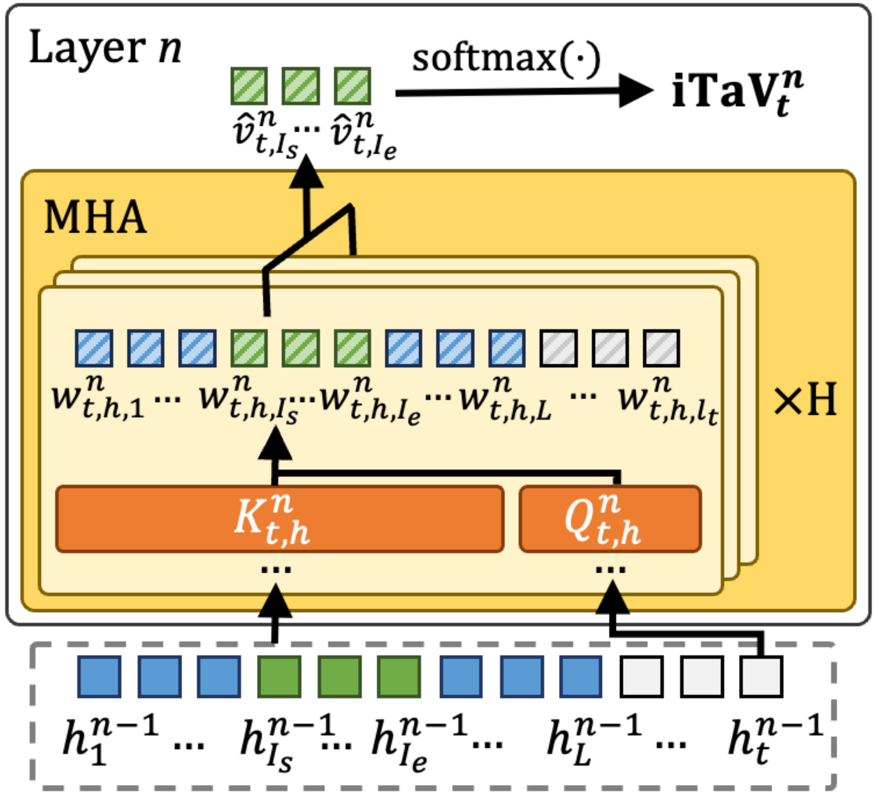
$$\mathbf{w}^n_{t,h} = [w^n_{t,h,1}, w^n_{t,h,2}, \ldots, w^n_{t,h,l_t}] = \texttt{softmax}\left(\frac{Q^n_{t,h} K^n_{t,h}}{\sqrt{d_k}}\right)$$

Selected maximum weight for each step:

$$\hat{v}^n_{t,j} = \max_{h=1,2,\ldots,H} w^n_{t,h,j}, \quad j = 1, 2, \ldots, l_t$$

Image token attention vector (iTaV):

$$\mathbf{iTaV}^n_t = \texttt{softmax}([\hat{v}^n_{t,I_s}, \hat{v}^n_{t,I_s+1}, \ldots, \hat{v}^n_{t,I_e}])$$



Distance Measurement: JS Divergence

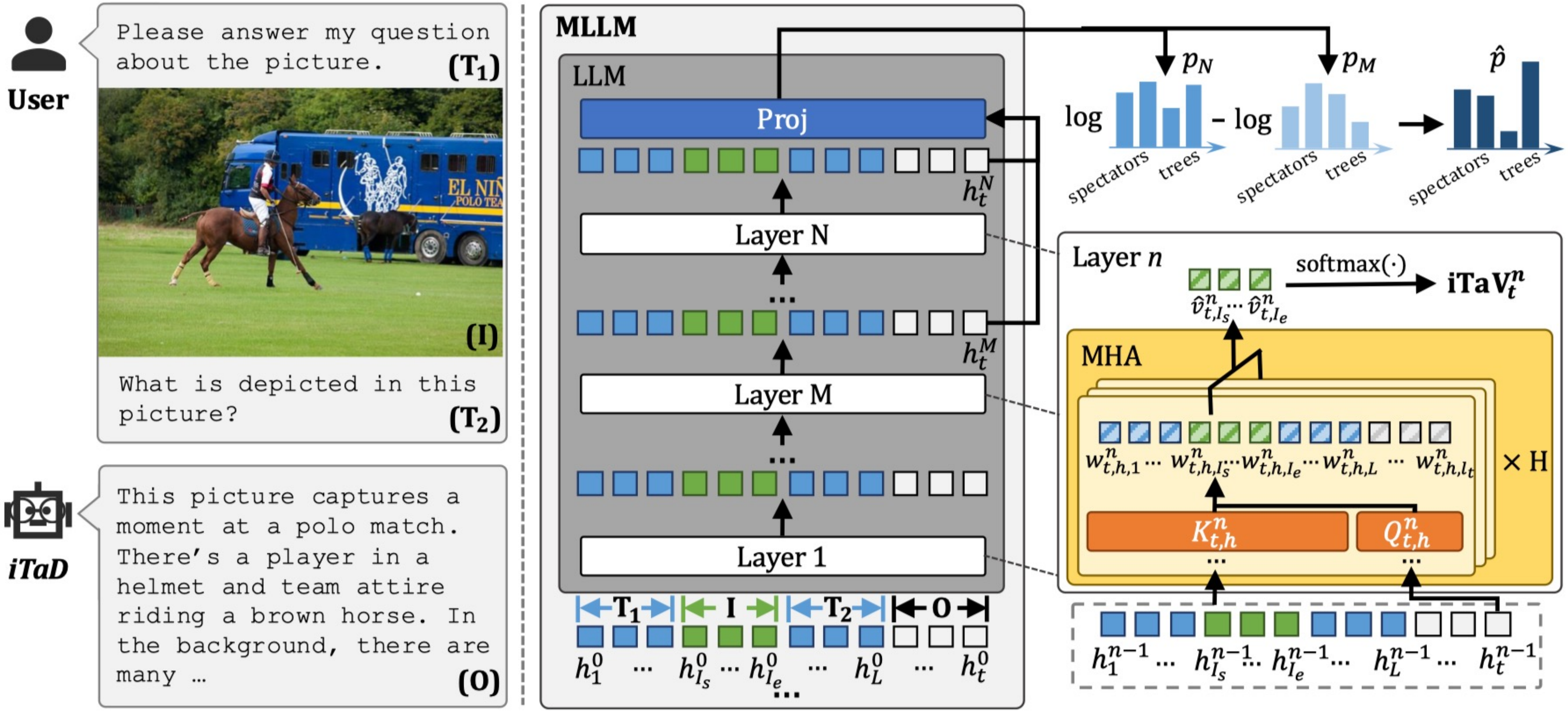
$$\text{dist}(\mathbf{iTaV}^i_t, \mathbf{iTaV}^j_t) = \text{JSD}(\mathbf{iTaV}^i_t || \mathbf{iTaV}^j_t)$$

iTaD: An Implementation

Intermediate Layer Selection Strategy

$$M = \max_{j \in \mathcal{M}} \mathrm{dist}(\mathbf{iTaV}_t^j, \mathbf{iTaV}_t^N)$$

The layer M with **the most distant iTaV** from the last layer N



An Overall Pipeline of iTaD

Main Results: CHAIR/POPE/GPT-4V Benchmark

CHAIR

Method	LLaVA-1.5				InstructBLIP				MiniGPT-4				mPLUG-Owl			
	$max\_l = 512$		$max\_l = 64$		$max\_l = 512$		$max\_l = 64$		$max\_l = 512$		$max\_l = 64$		$max\_l = 512$		$max\_l = 64$	
	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$
Greedy	47.8	13.7	20.4	6.7	57.3	23.9	29.1	13.5	31.8	10.7	22.0	7.6	71.0	24.8	33.4	13.0
Nucleus	52.2	15.9	24.9	8.5	56.2	25.6	30.2	14.8	33.6	11.2	23.4	8.5	76.6	28.1	38.1	16.0
Beam Search	50.0	14.5	19.4	6.3	56.4	16.2	23.6	8.0	33.9	10.7	23.2	7.9	75.8	25.4	31.4	**12.1**
DoLa	47.5	13.7	19.3	6.4	56.4	16.1	23.8	7.8	32.5	10.1	23.4	8.0	73.0	24.7	31.8	12.7
OPERA	47.2	13.5	19.1	6.5	53.4	15.3	24.0	**6.2**	27.8	9.7	21.8	7.7	73.8	25.1	31.1	12.4
iTaD	**45.4**	**13.4**	**19.0**	**6.2**	**53.2**	**14.7**	**22.2**	7.5	**26.4**	**9.6**	**20.7**	**7.5**	**70.0**	**24.5**	**29.5**	12.4

POPE

Method	LLaVA-1.5	InstructBLIP	MiniGPT-4	mPLUG-Owl
Greedy	82.2	80.0	58.5	68.5
Nucleus	82.5	80.1	57.8	70.1
Beam Search	84.9	84.4	70.3	69.2
DoLa	83.2	83.4	72.8	68.8
OPERA	85.4	84.8	73.3	67.6
iTaD	**85.5**	**85.2**	**75.5**	**72.3**

GPT-4V Assisted Evaluation

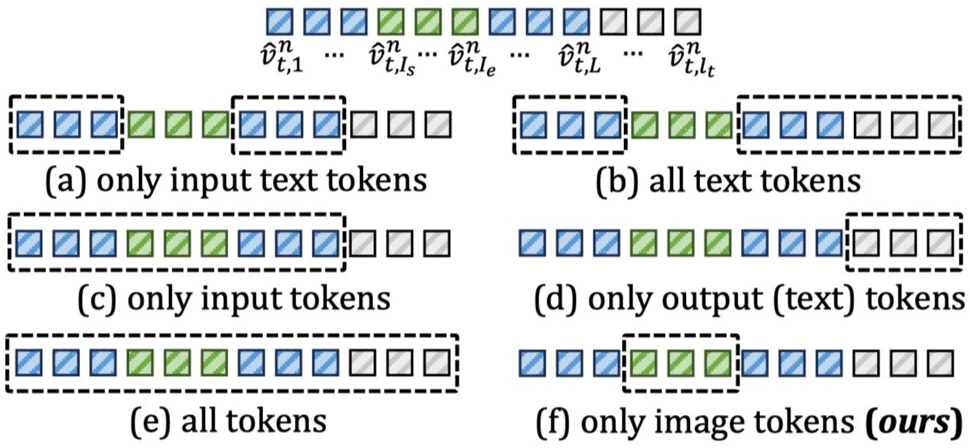
Method	LLaVA-1.5		InstructBLIP		MiniGPT-4		mPLUG-Owl	
	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$
Beam Search	5.7	5.5	4.9	**4.5**	5.4	**5.1**	4.2	4.5
iTaD	**6.8**	**5.7**	**5.5**	4.4	**6.5**	**5.1**	**5.2**	**4.6**

Method	LLaVA-1.5		InstructBLIP		MiniGPT-4		mPLUG-Owl	
	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$	$C \uparrow$	$D \uparrow$
OPERA	5.4	**5.3**	5.0	**4.9**	5.4	5.1	4.4	**4.3**
iTaD	**5.9**	**5.3**	**5.7**	**4.9**	**6.7**	**5.6**	**5.0**	4.1

iTaD outperforms the baselines in most cases.

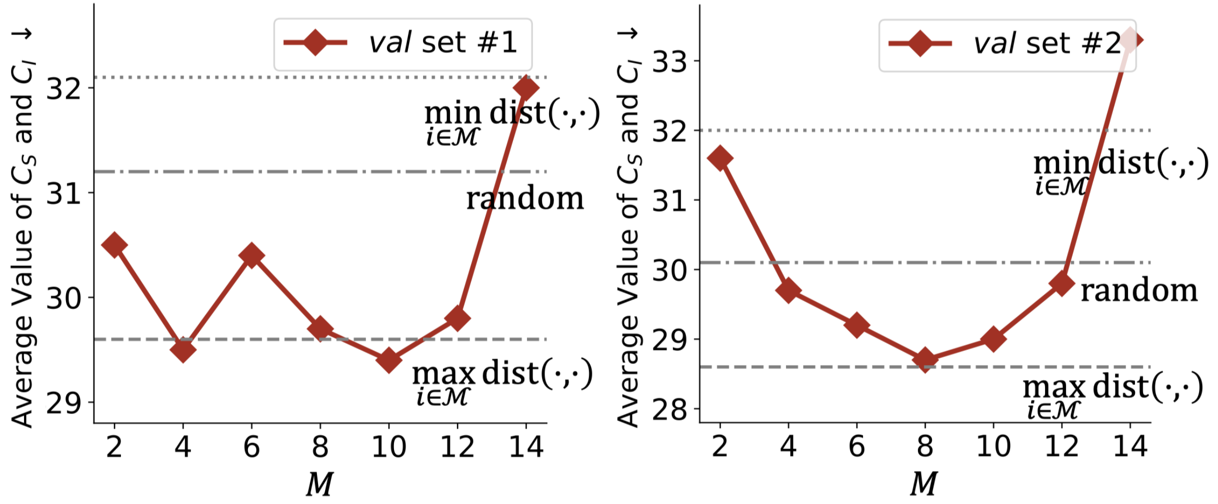
Analysis: Ablation Studies

Ablation Studies on iTaV



$\mathbf{iTaV}^n_t = \texttt{softmax}(\cdot)$	LLaVA-1.5		InstructBLIP		Avg.↓
	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	
(a) $[\hat{v}^n_{t,1}, ..., \hat{v}^n_{t,I_s-1}, \hat{v}^n_{t,I_e+1}, ..., \hat{v}^n_{t,L}]$	46.5	13.5	54.1	15.1	32.3
(b) $[\hat{v}^n_{t,1}, ..., \hat{v}^n_{t,I_s-1}, \hat{v}^n_{t,I_e+1}, ..., \hat{v}^n_{t,l_t}]$	<u>45.5</u>	13.6	<u>53.3</u>	14.8	<u>31.8</u>
(c) $[\hat{v}^n_{t,1}, ..., \hat{v}^n_{t,L}]$	46.3	13.7	53.4	**14.6**	32.0
(d) $[\hat{v}^n_{t,L+1}, ..., \hat{v}^n_{t,l_t}]$	47.4	14.0	54.3	15.5	32.8
(e) $[\hat{v}^n_{t,1}, ..., \hat{v}^n_{t,l_t}]$	46.2	**13.2**	53.8	14.8	32.0
(f) $[\hat{v}^n_{t,I_s}, ..., \hat{v}^n_{t,I_e}]$ ***(ours)***	**45.4**	<u>13.4</u>	**53.2**	<u>14.7</u>	**31.7**

Ablation Studies on the Selection of *M*



iTaD's selection strategy shows consistently superior performance across different validation sets.

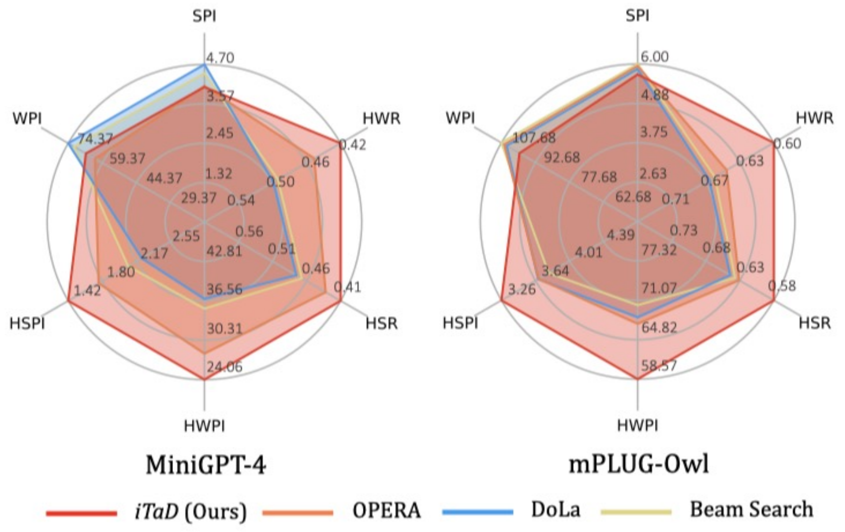
Analysis: Generalization and Latency

Cross-Dataset Validation

MME Benchmark

Model	Beam Search	OPERA	*iTaD*
MiniGPT-4	745.9	<u>757.4</u>	**772.3**
mPLUG-Owl	<u>1189.4</u>	1175.0	**1259.7**

GPT-4 Assisted Evaluation



Inference Latency (ms/token)

Method	Beam Search	OPERA	*iTaD*
LLaVA-1.5	56.0 ($\times$1.00)	283.4 ($\times$5.06)	60.7 ($\times$1.08)
InstructBLIP	33.2 ($\times$1.00)	190.1 ($\times$5.73)	37.7 ($\times$1.12)
MiniGPT-4	34.3 ($\times$1.00)	206.6 ($\times$6.02)	39.6 ($\times$1.15)
mPLUG-Owl	33.7 ($\times$1.00)	200.1 ($\times$5.94)	39.8 ($\times$1.18)

iTaD is robust across various datasets and can mitigate hallucinations with only minimal increase in latency.

Summary

- An observation that MLLM hallucinations often occur with output tokens' attention reduction to image tokens.
- A concept of the **image token attention vector** and its distance measurement to quantify the inter-layer difference in image understanding.
- An **intermediate layer selection strategy** for inter-layer contrastive decoding which exploits attention to image tokens and selects the layer with the most distant iTaV from the last layer.
- Effective across various benchmarks, models, datasets, outperforming the baselines.

Thank You!

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http://ise.thss.tsinghua.edu.cn/MIG/index.html